

AmbuRouteAI

 An AI-Powered Traffic Management for Emergency Vehicles.

 AmbuRouteAI is an AI-powered traffic management system that detects ambulances in real-time and dynamically controls traffic signals to provide a clear path for emergency vehicles. Using YOLOv8 for object detection and OpenCV for traffic light simulation, this system helps reduce ambulance response time and enhances emergency healthcare logistics.

Features

✅ Real-time Ambulance Detection – Uses YOLOv8 AI to identify ambulances from live traffic feeds. ✅
Smart Traffic Signal Control – Dynamically changes traffic lights using OpenCV when an ambulance is detected. ✅
Live Video Processing – Works with webcams or traffic camera feeds to analyze road conditions. ✅
Seamless Integration – Can be extended to IoT-enabled smart city infrastructure. ✅
Scalable & Efficient – Future-ready for integration with Google Maps API for real-time route optimization.

Tech Stack

Component Technology/Tool
AI Model YOLOv8 (Ultralytics) Computer Vision OpenCV, NumPy Backend
Python Traffic Simulation OpenCV Live Video Input Webcam / CCTV Feed

How It Works

1 AI-Powered Detection – YOLOv8 identifies ambulances in real-time from traffic video feeds. 2 Traffic Light Control – If an ambulance is detected, the traffic light turns green; otherwise, it remains red. 3 Visual Alerts & UI – The system draws bounding boxes around detected ambulances and simulates traffic signals. 4 Extensibility – The project can integrate with Google Maps API & IoT sensors for smarter city-wide traffic control.

Project Structure

 AmbuRouteAI |  README.md # Project documentation |  requirements.txt # Required dependencies |  main.py # Main Python script |  models # YOLO model files |  media # Sample images/videos |  datasets # Training datasets (optional)

Installation & Setup

1 Clone the Repository

git clone <https://github.com/mantreshkhurana/AmbuRouteAI.git> cd AmbuRouteAI

2 Install Dependencies

```
pip install -r requirements.txt
```

3 Download YOLOv8 Model

Download the pre-trained YOLOv8 model from Ultralytics and place it in the models folder.

```
wget https://github.com/ultralytics/assets/releases/download/v0.0.0/yolov8n.pt -P models/
```

4 Run the Project

To test with a sample traffic video:

```
python main.py
```

To run with live webcam:

```
python main.py --source 0
```

🎯 Future Enhancements

◆ Integrate with Google Maps API – Real-time traffic data for route optimization. ◆ IoT-Based Smart Traffic Signals – Connect with Raspberry Pi & Arduino for real-world applications. ◆ Hospital Alert System – Send emergency notifications to hospitals about incoming patients. ◆ Mobile App Interface – Develop an ambulance tracking app with live traffic updates.

🖼️ Demo & Screenshots

📌 Detecting an ambulance and switching the traffic light in real-time.

Authors

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